



Seabirds have tiny glands in the nasal cavity that collect the salt, which they then shake or sneeze out.

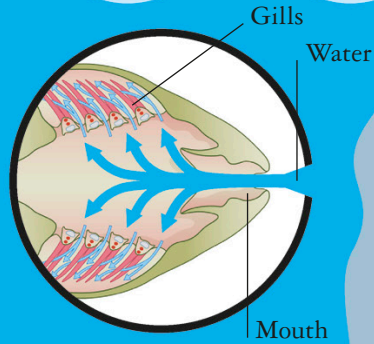
BARNACLES

Just try getting me off this rock!



BREATHING

Like land animals, marine creatures breathe oxygen. Mammals and reptiles have lungs, so they have to come to the surface at intervals to take a gulp of air. They are good at holding their breath and diverting blood away from nonessential areas, such as the flippers, so that more oxygen goes to the heart and brain.



Fish have gills on either side of their head. Blood vessels here filter out the oxygen from seawater.

Rocky shores

Organisms that live close to the shore may not have to cope with pressure or lack of light, but they do have other challenges. Invertebrates and plants are constantly pounded by waves and they have to anchor themselves securely to a rock. Other animals are left exposed when the tide goes out, and have a protective shell that they can seal to prevent themselves drying out.

However, most underwater creatures have to get oxygen from the water rather than the air. Fish and invertebrates that spend all their time submerged have gills or respire through their skin to extract oxygen from water flowing over them.

For every 33 ft (10 m) below the surface, the *pressure* **INCREASES** by the equivalent of one atmosphere.

HELP!

They say there's a lot of pressure on those who live down here...

... but since we don't have lungs, it doesn't bother us at all.

Under pressure

Although we can't feel it, on dry land we have the weight of the atmosphere pressing down on every square inch of our bodies. Dive into the ocean and you also have the weight of the water pressing down. The deeper you go, the more the pressure increases and squashes the airspaces of animals with lungs. Instead of resisting the pressure, deep divers such as sperm whales and elephant seals can collapse their lungs, slow their heartbeat, and store oxygen in their muscles. This also helps them to sink, so they don't spend as much energy swimming. If human scuba divers return to the surface too quickly they can die from the rapid change in pressure, which causes bubbles to form in the blood.